

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, m, i, j, k;
```

```
    int alloc[10][10], max[10][10], need[10][10];
```

```
    int avail[10], finish[10] = {0}, safe[10];
```

```
    printf("Enter number of processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter number of resources: ");
```

```
    scanf("%d", &m);
```

```
    printf("Enter Allocation Matrix:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        for(j = 0; j < m; j++)
```

```
            scanf("%d", &alloc[i][j]);
```

```
    printf("Enter Maximum Matrix:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        for(j = 0; j < m; j++)
```

```
            scanf("%d", &max[i][j]);
```

```
    printf("Enter Available Resources:\n");
```

```
    for(i = 0; i < m; i++)
```

```
        scanf("%d", &avail[i]);
```

```
for(i = 0; i < n; i++)  
    for(j = 0; j < m; j++)  
        need[i][j] = max[i][j] - alloc[i][j];
```

```
int count = 0;
```

```
while(count < n) {  
    for(i = 0; i < n; i++) {  
        if(finish[i] == 0) {  
            for(j = 0; j < m; j++)  
                if(need[i][j] > avail[j])  
                    break;  
  
            if(j == m) {  
                safe[count++] = i;  
                finish[i] = 1;  
  
                for(k = 0; k < m; k++)  
                    avail[k] += alloc[i][k];  
            }  
        }  
    }  
}
```

```
printf("Safe Sequence: ");
```

```
for(i = 0; i < n; i++)  
    printf("P%d ", safe[i]);  
  
return 0;  
}
```